Data Analytics (DA)

Module-End -5

Module End Assignment: Python For Data Analysis

“Social Media Engagement Analytics”

Assignment Task Overview

Project Title: Social Media Engagement Analytics Using Python

Problem Statement

Social media platforms generate massive volumes of engagement data—likes, comments, shares, impressions, watch time, and more. Analyzing such data helps companies understand user behavior, identify trends, and improve content performance.

In this assignment, you will work with a **social media dataset** containing social media engagement metrics. The task involves **data cleaning, transformation, NumPy/Pandas operations, exploratory data analysis, visualizations**, and generating insights.

Dataset Provided:

social_media_engagement_5000.csv

Assignment Tasks

- **Task 1 — Data Import & Setup**
  - Import CSV using Pandas
  - Check/convert data types
  - Convert date columns to datetime
  - Use NumPy arrays for basic array operations
- **Task 2 — Data Cleaning**
  - ***Cleaning Missing Data***
    - Detect missing values (isnull(), isna())
    - Handle using: dropna(), fillna(), median/mode, forward/backward-fill
  - ***Duplicate Handling***
    - Identify & remove duplicates

- ***Data Formatting***
  - Fix incorrect data types
  - Standardize categories (e.g., gender labels)
  - Correct unrealistic values in likes, comments, shares
- ***Feature Cleaning***
  - Extract hashtag count
  - Clean sentiment labels

❖ Task 3 — Data Exploration using Pandas

Perform the following:

- View dataset structure using `head()`, `tail()`, `shape`, and `columns`.
- Check data types and info with `info()` and `dtypes`.
- Generate summary statistics using `describe()`.
- Analyze categorical distributions using `value_counts()`, `unique()`, and `nunique()`.
- Create a correlation matrix for numeric fields.
- Use `groupby()` to summarize metrics (e.g., avg likes by post type, impressions by country).

❖ Task 4 — Data Wrangling

- Use **merge**, **concat**, or **join** if combining DataFrames.
- Create new fields such as **engagement_score**, **log-transformed metrics** (optional), and **hashtag_count**.
- Perform **groupby** summaries by **post_type**, **country**, and **sentiment**.

❖ Task 5 — Statistical Analysis

Compute descriptive stats for **likes, comments, shares, watch_time, engagement_rate, followers**:

- **Mean, median, mode**
- **Standard deviation, variance**
- **Percentiles**
- *(Optional)* **Skewness and kurtosis**

❖ Task 6 — Data Visualization (Min. 8 Plots Required)

- **Matplotlib**
  - **Scatter:** *likes vs impressions*
  - **Line:** *daily engagement trend*
  - **Bar:** *posts by category*
  - **Pie:** *gender distribution*
  - **Histogram:** *age*
  - **Box:** *engagement rate*

- **Seaborn**
  - **Count plot:** *post type*
  - **Bar plot:** *avg likes by category*
  - **Violin:** *followers vs sentiment*
  - **Pair plot:** *numeric features*
  - **Heatmap:** *correlation matrix*
  - **Swarm plot:** *engagement vs device*
- **Plotly (Interactive)**
  - *Interactive line chart/bar chart/bubble/scatter chart*

❖ Final Insights should Include following analysis

- **Content Performance**
  - *Which post types have highest engagement?*
  - *Best-performing content category?*
  - *Which countries have highest avg engagement rate?*
- **User Trends**
  - *How age affects engagement*
  - *Performance difference for verified accounts*

Behavioral Insights

- **Best time of day for impressions**

- **Device type impact on watch time**

Sentiment Analysis

- **Which sentiment performs best**
- **Behavior of negative/neutral sentiment posts**

Deliverables:

- Complete the assignment in Google Colab or Jupyter Notebook.
- For Google Colab, share the notebook link with "View" access. If using Jupyter, upload the notebook to Google Drive and ensure "View" access is enabled.
- Clear one-page summary with findings supported by analysis

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Expected Proficiency
Data Loading & Organization	Python Basics	Lists, Dictionaries, Appending Data	Beginner
Data Cleaning & Transformation	Text Processing	Lowercasing, punctuation removal, spacing	Intermediate
Keyword & Text Analysis	Logical/Analytical Skills	Word search, counting, string operations	Intermediate

Summary Statistics	Data Analysis	Average rating, longest text, unique words	Intermediate
Data Structuring & Sorting	Data Handling	Sorting with zip(), filtering, restructuring	Intermediate
Reporting & Insights	Communication	Clear summary of findings & patterns	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill/Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Credit (Rounded Range)
Data Loading & Setup	3	All ticket data correctly loaded, structured, and appended	Minor issues in loading or formatting	**1–2**
Data Cleaning & Text Processing	5	All cleaning steps completed accurately (lowercase, punctuation removal, spacing, slang fix)	Some steps correct, but partially incomplete or inconsistent	**3–4**

Functions, Loops & Logic	4	All required functions work correctly (word count, loops, conditionals)	Functions partially correct or missing edge cases	**2–3**
Summary Statistics	4	Accurate average rating, longest feedback, unique word extraction	Most calculations correct but incomplete	**2–3**
Sorting & Structuring Output	3	Correct ranking/sorting of tickets and structured display	Partially sorted or formatting errors	**1–2**
Output Presentation	4	Clear, readable outputs in notebook (tables/lists well formatted)	Outputs present but not clearly formatted	**2–3**
Reporting & Insights	2	Clear one-page summary with findings supported by analysis	Summary incomplete or lacks clarity	**1**

Skill Taxonomy Structure

- **Data Loading & Management**
  - ➢ Loading dictionaries/lists
  - ➢ Adding new entries
  - ➢ Structuring data properly
- **Data Cleaning**
  - ➢ Lowercasing, punctuation removal
  - ➢ Space normalization
  - ➢ Slang/keyword corrections
  - ➢ Consistency checks

- **Functions & Analysis**
  - ➢ User-defined functions
  - ➢ Loops, conditionals
  - ➢ Word counting, logical operations
  - ➢ Basic statistical measures
- **Data Summarization**
  - ➢ Sorting data
  - ➢ Filtering and grouping
  - ➢ Extracting unique words

Evaluation Guidelines

1. Review the complete code, including code, outputs, and explanations.
2. For each section, match student work to the specific skill and sub-skill area in the taxonomy.
3. Assign full or partial marks per the rubric, using suggested partial mark ranges for "partially complete" work.
4. For ambiguous or incomplete submissions, note what's missing and correspondingly reduce credit.
5. Comments must explain strengths and improvement points for each section.
6. Objective assessment: focus on function, correctness, clarity, not on visual design beyond the stated requirements.

Achievement Criteria & Passing Rules

- ➢ At least partial (not zero) credit must be received in every section—no component can be left unattempted.
- ➢ Students must fully achieve at least two major skills**: Data Cleaning & Text Processing,Functions, Loops & Logic and Summary Statistics(min,max,count..)**
- ➢ Feedback and a detailed scorecard must be given for all evaluated assignments.